Ch. 4 - Fisherian Inference and ML Estimation LL

Likelihood and Maximum Likelihood

$$l_x(\mu) = \log\{f_\mu(x)\}$$

MLE: $\hat{\mu} = \underset{\mu \in \Omega}{\arg\max}\{l_x(\mu)\}$

Iconic method b/c ...

1.

2.

3.

Many MLEs require numerical ________.

MLEs for individual parameters may work well but the set of MLEs for a large # of parameters may be misleading. Leads to regularization concept.

Fisher Information and MLE

Consider a family of densities, $\mathcal{F} = \{f_\theta(x), \theta \in \Omega, x \in \mathcal{X}\}$ with $P(A) = \int_A f_\theta(x)\,dx$.

Then $l_x(\theta) = \log f_\theta(x)$ and MLE $\hat{\theta} = \arg\max\{l_x(\theta)\}$

The score fn is defined as $\dot{l}_x(\theta) = \frac{\partial}{\partial\theta}\log f_\theta(x)$

The score fn has expectation 0 (proof in text).

Fisher Information is defined as the variance of the score fn.

$$I_\theta = \int_x \dot{l}_x(\theta)^2 f_\theta(x)\,dx$$

It turns out MLE has $\approx N(\theta, \overset{\text{Var}}{1/I_\theta})$, and no nearly unbiased est. of θ can do better.

For $x = (x_1, x_2, \ldots, x_n)$ iid, you can derive Fisher's Fundamental Thm for MLE, which is

$$\hat{\theta}_n \;\dot{\sim}\; N(\theta, 1/(nI_\theta)). \quad \text{(for large } n\text{)}$$

Normal ex. in text. X's iid $N(\theta, \sigma^2)$.

$\hat{\theta} = \bar{x}$ and $I_\theta = \frac{1}{\sigma^2} \Rightarrow \hat{\theta} \sim N(\quad , \quad)$

<u>How to compute Fisher Info?</u>

Don't use the definition. There is an easier computational formula.

Def. $I(\theta) = E_\theta \left\{ \left[\frac{\partial}{\partial\theta} \log f(x|\theta) \right]^2 \right\}$

Comp. formula $I(\theta) = -E_\theta \left[\frac{\partial^2}{\partial\theta^2} \log f(x|\theta) \right]$

For a random sample, $I_n(\theta) = nI(\theta)$.
(Used in Fisher's Fundamental Thm).

Cramer-Rao. L3

Suppose $\tilde{\theta} = t(x)$ is unbiased for θ based on n iid obs. Then the Cramer-Rao lower bound for the variance of $\tilde{\theta}$ is

$$\text{Var}_\theta\{\tilde{\theta}\} \geq \frac{1}{nI_\theta}.$$

$\Longrightarrow$ Efficient estimators

Practice. Suppose $X_1, \dots, X_n$ is RS from Poisson (θ), ie. $f(x|\theta) = \dfrac{e^{-\theta}\theta^x}{x!}$, $x = 0, 1, 2, \dots$ and $\theta > 0$.

Find MLE for θ. Verify the MLE is unbiased.

Find $I(\theta)$ (for a single obs.).

Find the Cramer-Rao lower bound. Does the MLE achieve the lower bound on variance?

Ch. 4 concludes with topics of Conditional Inference and Permutation and Randomization. Be sure you understand how to run permutation-based procedures, such as a 2 sample permutation t-test.

Ch. 5 Parametric Models and Exponential Families

Why parametric models? ________________

Reviews several distributions

5.2 Multivariate Normal (Linear Algebra is useful!)

$x = (x_1, x_2, \ldots, x_p)'$ is a random vector

$x \sim N(\mu, \Sigma)$ μ is mean vector
Σ is ____________ matrix.

Key results for conditional dists, including the univariate case. Connection to Bayesian applications. For the univariate case, $\mu \sim N(M, A)$ and $x|\mu \sim N(\mu, \sigma^2)$ then

$$\mu | x \sim N\left(M + \frac{A}{A+\sigma^2}(x - M), \frac{A\sigma^2}{A+\sigma^2}\right).$$

The posterior mean ↑ is a shrinkage estimator.

5.3 Fisher Info → Properties extend. Need more calc and linear algebra.

Key result: The variance of the MLE must always ____________ in the presence of nuisance parameters.

5.4 Multinomial Dist. n cases

$x = (x_1, x_2, \ldots, x_L)$ with probs $\pi = (\pi_1, \pi_2, \ldots, \pi_L)$.

$$x \sim \mathrm{Mult}_L(n, \pi)$$

Connection to Poisson. Suppose $S_1, S_2, \ldots, S_L$ are $\perp$ Poissons with $S_\ell \sim \mathrm{Poisson}(\mu_\ell)$, $\ell = 1, \ldots, L$ (Conditional diff.)

Then the collection $S = (S_1, S_2, \ldots, S_L)$ has a cond. dist. given the sum $S_+ = \sum S_\ell$ that is Multinomial.

i.e. $S \mid S_+ \sim \mathrm{Mult}_L(S_+, \mu/\mu_+)$, where $\mu_+ = \sum \mu_\ell$.

If $N \sim \mathrm{Poisson}(n)$ then the marginal dist of $\mathrm{Mult}_L(N, \pi)$ is Poisson. $\mathrm{Mult}_L(N, \pi) \sim \mathrm{Poisson}(n\pi)$

5.5. Exponential Families

Characterized by form and behave very nicely.

Ex in text. Consider Poisson @ μ and μ_0. Take ratio: $f_\mu(x) / f_{\mu_0}(x) = e^{-(\mu - \mu_0)} \left(\frac{\mu}{\mu_0}\right)^x$. Re-express

$$f_\mu(x) = e^{\alpha x - \psi(\alpha)} f_{\mu_0}(x), \text{ where}$$

$$\alpha = \log(\mu/\mu_0) \text{ and } \psi(\alpha) = \mu_0 (e^\alpha - 1).$$

This is a renormalized exponential tilt.

Extends to p-parameter families, like Gamma or Beta.

$\psi(\alpha)$ is called the cumulant generating fn for 1-parameter families. Connections to sufficient stats. Will see another way to consider this in problems.